

Business Management Systems

Intellectual Property

Riaz Esmailzadeh

We respectfully acknowledge the Kaurna, Boandik, and Barngarla First Nations Peoples and their Elders past and present, who are the Traditional Owners of the lands that are home to our campuses across Adelaide and South Australia.

Outline

- Definitions
- An inventor's journey
- A career as a patent attorney

Intellectual Property

- According to business.gov.au, the government website for the Australian business community:
 - “Intellectual property (IP) is a creation of your mind or exclusive knowledge. Any new products, services, processes or ideas you develop are your IP.”

IP Rights Classification

- Again, according to business.gov.au there are many IP protection rights:
 - “Patents: which protect inventions and new processes
 - Trade marks: which protect logos, words and other branding
 - Copyright: which protects art, writing, music, film, and computer programs
 - Registered designs: which protect the visual design of a product
 - Circuit layout rights: which protect layout designs or plans of integrated circuits used in computer-generated designs
 - Plant breeder’s rights: which protect the commercial rights of new plant varieties.”

Patent

- Patents protect inventions (not ideas, but embodied ideas)
- Duration up to 20 years (different for drugs – 25 years)
- It can be acquired if:
 - The invention has a practical use
 - It is novel: no one has thought of it before
 - It is not obvious to someone practiced in the art
- Patent rights can be enforced through courts of law
 - If a non-authorized person does something that falls within what the patent claims

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Trade Mark ®



- A trade mark protects brands
 - Basically a sign which distinguishes a trader's goods/services.
- Duration of right is indefinite: even if the owning company goes out of business, they retain the right
- To acquire, one must have used the trademark in commerce before anyone else. (Although this can be hard to prove...)
- To enforce trade mark rights, you must prove in court that the new trade mark is “substantially identical and/or deceptively similar”



deckers
OUTDOOR CORPORATION

[Luda Corporation](#)

Copyright ©

- A Copyright protects an expression of an idea.
 - Literary, artistic, musical, performance
 - Also computer programs (in C++, Java, etc.)
- Copyright endures the life of the author plus 70 after his/her death
- Copyright is acquired automatically when it is recorded in a material form (no need to put a © on your work)
- If someone copies it substantially, then they are infringing and can be sued.

Land Down Under
Kookaburra sits in the Old Gum Tree
“Men at Work” had to pay 5% royalty

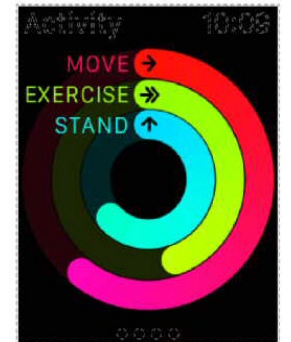


Registered Design

- Registered designs protects the appearance of a product
- Duration of design rights is 10-25 years
- To acquire one must file a design registration
 - The design cannot be identical or substantially similar to another product
- You can then enforce it in courts if another product is identical or substantially similar



AU design no.
342337



AU design
no. 360961



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Trade Secret



- Another way of protecting one's intellectual property is to keep it a secret
 - Example: Coke and KFC formulas, closed-source software
- The trade secret can be kept as long as well it is a secret!
- You can not apply for it in a patent office, since that will lead to disclosure
- You can share it with business partners under non-disclosure agreement (NDA)
 - If the recipient discloses, they can be sued in court
- If someone reverse engineers your formula, or comes up with something similar, you cannot do anything ineffective against reverse engineering or independent formulation

Why Patent

- Patent is basically a legal contract between the patentee and the government
- It gives you the right to exclude others from exploiting your invention for a fixed period
 - In return for a full disclosure of the invention
- This encourages R&D (e.g. pharmaceuticals) as you will have the exclusive right to enjoy the revenue an idea may generate
- It also provides the society with innovations which at a later date may be enjoyed by everyone.

Intangible Asset

- Patent (and other IP) is an intangible asset with the following benefit
 - Source of income – in exclusive sales and royalties through licensing
 - Protection of business by deterring competition
 - It may be sold
 - It can be the most valuable asset of a company



My Journey as an Inventor!



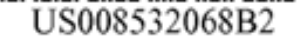
- I was lucky: right place at the right time...
- My PhD thesis was on Time Division Duplex – Code Division Multiple Access (TDD-CDMA) technology – first presented in 1991
 - My professor's idea (Prof Nakagawa at Keio University in Japan)
 - We were the first to propose the idea and develop its basic technologies
 - All patentable inventions which we never applied for...
 - Like most other PhD theses...
 - Except ours became the basis for a global 3G mobile standard (TD-CDMA)
 - HUGE MISTAKE!!!

“Wireless Internet”

- My first job out of Keio was at Hitachi (1994-96)
 - Researching technologies for 3G standards
 - Actually beyond 3G, as we looked at how mobile telephony could evolve to carry internet traffic
 - Second generation, *digital* mobile systems were just coming to market
 - E.g. GSM
 - I applied for my first patents at Hitachi
 - And actually made some money from them... (although not much upfront)



- The first page of a patent document shows a number of details
 - Inventor (me and two other colleagues) and the assignee – that is the owner – FIPA who had bought it from Hitachi
 - Priority date (and by extension the date it expires)
 - Supporting documents



(10) **Patent No.:** US 8,532,068 B2
(45) **Date of Patent:** *Sep. 10, 2013

10 Claims, 10 Drawing Sheets

Patent Sections

- There are three distinct parts to a patent
 - These give information intended to help a person practiced in the art to independently develop the product
- In US Patents the order is:
 - Drawings: illustrate the patent
 - Specification: are a detailed description of the patent, including prior art, novelty of the invention, and one or more ways of how it may be embodied
 - Claims: what the inventor claims to be novel, compared with the prior art
 - Let's have a look at the initial patent 6,111,869.

Drawings

- Figure 2 shows a telecom receiver that can embody the invention
- The ideas and how they work in a block diagram format are shown
- Note all components of the figure are numbered so they can be referred to in the specification

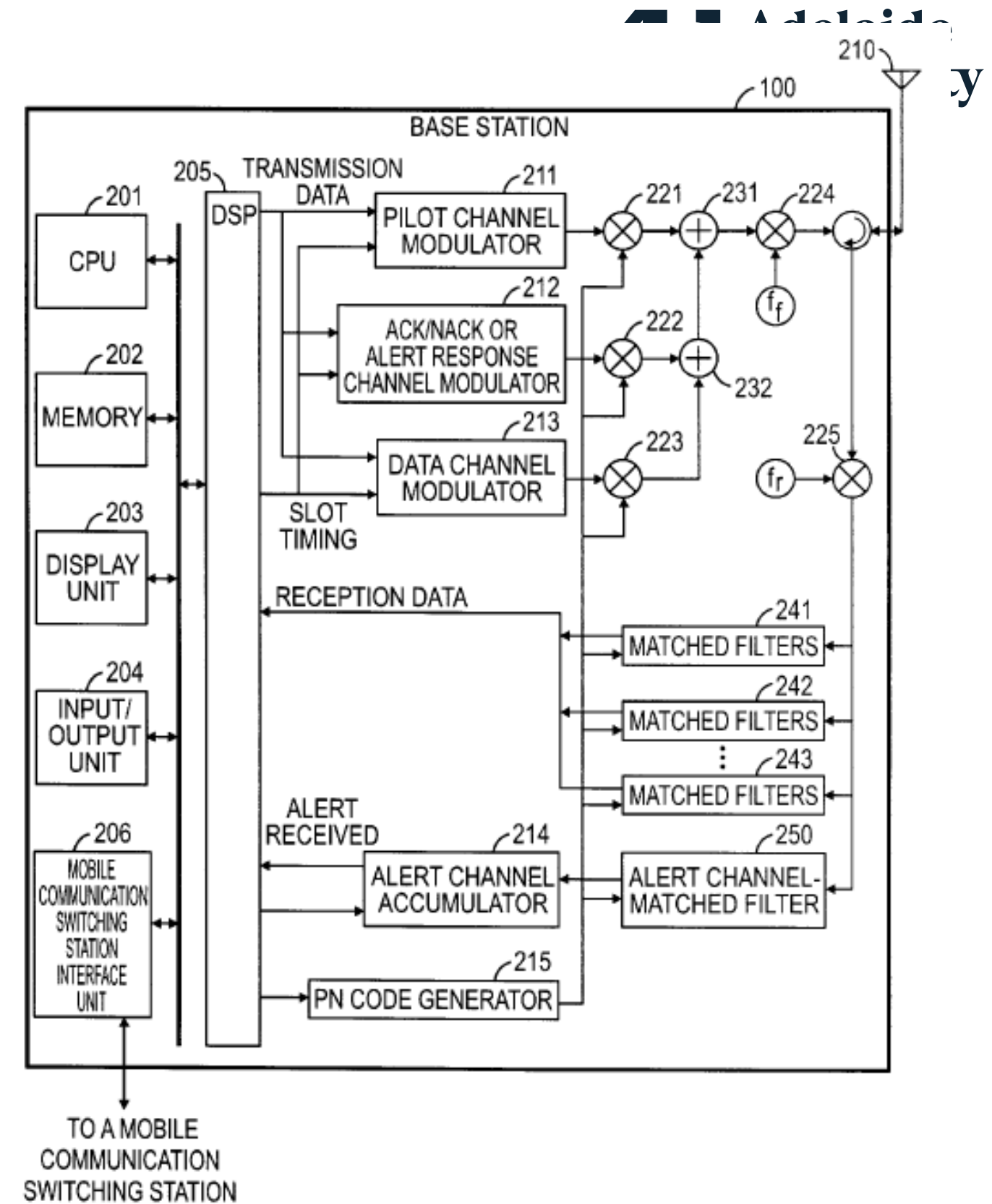


FIG. 2

Specification

FIELD OF THE INVENTION

The present invention relates to a method of mobile communication for transmitting and receiving data between a base station and a plurality of mobile terminals after having determined schedules for transmitting data, and to an apparatus therefor. Particularly, the invention relates to a method of mobile communication wherein, in transmitting a

BACKGROUND OF THE INVENTION

In the prior art, a reservation-type mobile communication system has been proposed to transmit and receive data by radio communication between a base station and a plurality of mobile terminals after having reserved data channels for transmission and reception.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

The following detailed description of the preferred embodiments is undertaken in connection with the drawings.

A mobile communication method and an apparatus there-

FIG. 2 illustrates base station **100** in the mobile communication apparatus in more detail. Base station **100** includes CPU **201**, memory **202**, display unit **203**, input/output unit **204**, DSP **205**, mobile communication switching station interface unit **206**, antenna **210**, pilot channel modulator **211**, ACK/NACK or alert response channel modulator **212**,

The effects obtained by the disclosed embodiment of the present invention are briefly set forth as follows.

(1) The data are transmitted and received after having determined the schedules by alert signals of orthogonal codes. Therefore, efficient communication of data is accomplished between the base station and the plurality of mobile terminals.

Claims

- The claims section is what really matters
- This is what the invention covers
- It specifies the scope of the invention, what it covers and therefore what a competitor may or may not do
- I believe this patent covers a step within 3G standards
 - Therefore, anyone making a 3G device needed to license it.

What is claimed is:

1. A method of mobile communication by determining schedules and by transmitting and receiving first data between a base station and a plurality of mobile terminals according to the schedules, the method comprising the steps of:

transmitting a particular orthogonal code selected from a plurality of orthogonal codes stored in a table from a mobile terminal to the base station as an alert signal indicating the presence of first data to be transmitted; detecting, at the base station, whether the particular orthogonal code is contained in the alert signals transmitted from the mobile terminals to the base station; when the particular orthogonal code is detected in the alert signal transmitted from the mobile stations to the base station, second data representing the detected orthogonal code and a schedule for transmitting the first data from the mobile terminal to the base station is transmitted from the base station to the mobile terminal as an alert response to the alert signal; and when the particular orthogonal code used for the alert signal transmitted from the mobile terminal to the base station is contained in the alert response transmitted from the base station to the mobile terminal, the schedule for transmitting the first data is read at the mobile terminal and the first data is transmitted from the mobile terminal to the base station accordingly.

Random Access

- After Hitachi I continued my work on 3G standards at Ericsson
- Here again, I was lucky to be at a right place at a right time.
- One of my inventions became an essential part of 3G standards
 - And generated a LOT of money for ... Ericsson


 US006259724B1

(12) **United States Patent**
Esmailzadeh

(54) **RANDOM ACCESS IN A MOBILE TELECOMMUNICATIONS SYSTEM**

(75) Inventor: **Riaz Esmailzadeh**, Stockholm (SE)

(73) Assignee: **Telefonaktiebolaget L M Ericsson (PUBL)** (SE)

(*) Notice: This patent issued on a continued prosecution application filed under 37 CFR 1.53(d), and is subject to the twenty year patent term provisions of 35 U.S.C. 154(a)(2).

Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **08/733,501**

(22) Filed: **Oct. 18, 1996**

(10) Patent No.: **US 6,259,724 B1**

(45) Date of Patent: ***Jul. 10, 2001**

0 257 110 A1 8/1986 (EP) .
 0 633 671 A2 5/1994 (EP) .
 WO 92/21196 11/1992 (WO) .
 WO 95/14336 5/1995 (WO) .

OTHER PUBLICATIONS

Design Study for a CDMA-Based Third-Generation Mobile Radio System by Alfred Baier, Senior Member, IEEE, Uwe-Carsten Fiebig, Member, IEEE, Wolfgang Granzow, Wolfgang Koch, Member, IEEE, Paul Teder, and Jörn Thielecke, Member, IEEE IEEE Journal on Selected Areas in Communications, vol. 12, No. 4, May 1994.
 Design Study for a CDMA-Based Third-Generation Mobile Radio System IEEE JSAC Special Issue on "CDMA Networks", Dec. 16, 1993.

(List continued on next page.)

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How Much Money?!

- Just as one illustration, Ericsson sued Samsung for patent infringement in 2012
 - A collection of 13 patents (the '724 the first on the list)
 - Civil Action No. 6:12-cv-895, in the United States District Court for the Eastern District of Texas Tyler Division ([link](#) page 8).
- It was settled in January 2014 when Samsung paid Ericsson US\$650 million plus on-going royalties ([link](#)).
 - Not a cent to me though... (I must say they had paid me quite a bit upfront)

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THE PATENTS

21. United States Letters Patent No. 6,259,724 ("the '724 Patent"), entitled "Random Access in a Mobile Telecommunications System," was duly and legally issued to inventor Riaz Esmailzadeh on July 10, 2001. Ericsson owns by assignment the entire right, title, and interest in the '724 Patent, and is entitled to sue for past and future infringement.

My “YouTube” Patent

- I left Ericsson to co-found Genista a software company in Tokyo area in May 2000.
- We designed software that controlled communications quality for data services: Voice, Audio and Video (YouTube was launched in February 2005)
- Our patent portfolio grew quickly – we applied for some 15 patents
- One of these was a patent to adapt the speed of transmission to the quality of the communications channel



(19) World Intellectual Property Organization
International Bureau(43) International Publication Date
18 April 2002 (18.04.2002)

PCT

(10) International Publication Number
WO 02/32161 A2(51) International Patent Classification⁷: H04Q 7/00

(21) International Application Number: PCT/JP01/08984

(22) International Filing Date: 12 October 2001 (12.10.2001)

(25) Filing Language: English

(26) Publication Language: English

(30) Priority Data:
60/240,434 13 October 2000 (13.10.2000) US(71) Applicant (for all designated States except US):
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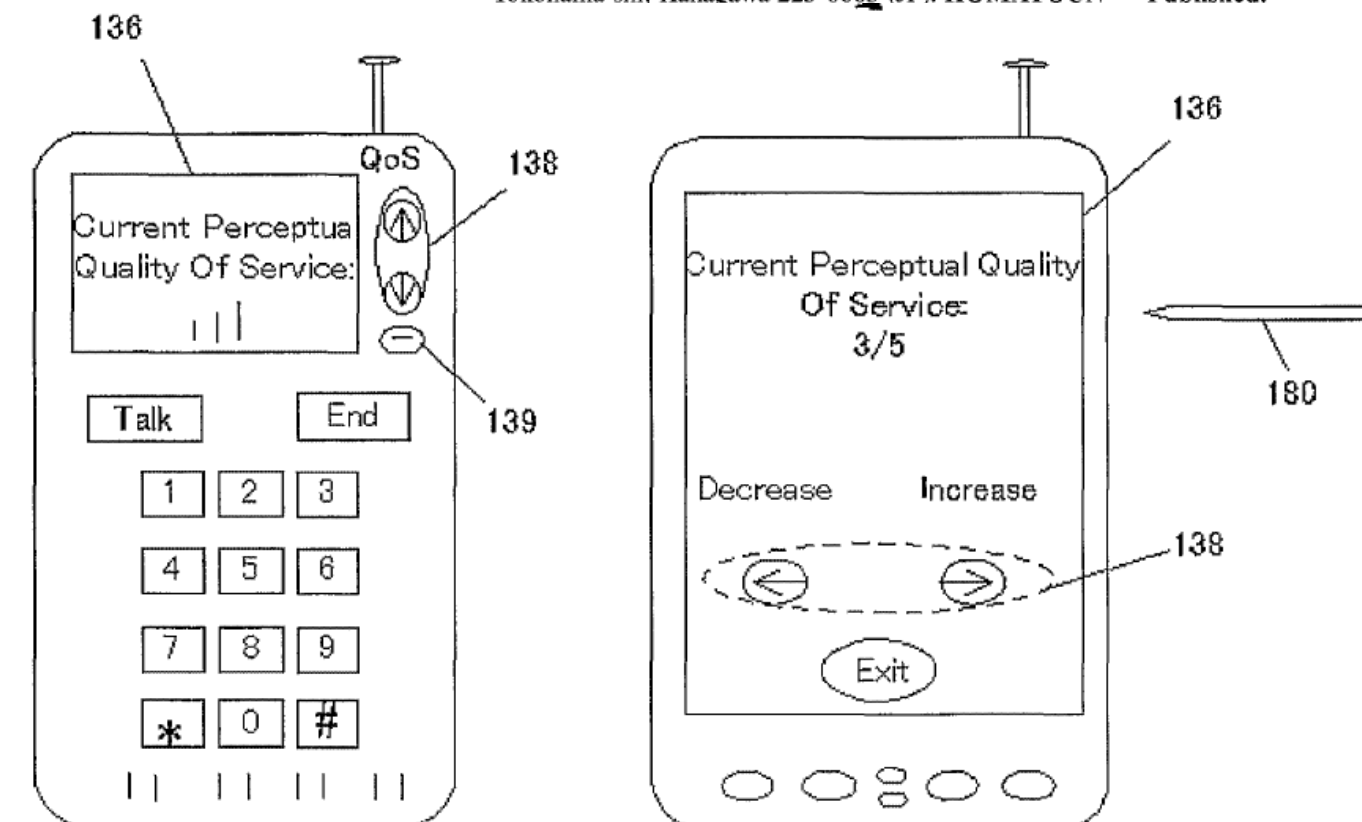
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105-0004 (JP).(81) Designated States (national): AE, AG, AL, AM, AT, AU,
AZ, BA, BB, BG, BR, BY, BZ, CA, CH, CN, CO, CR, CU,
CZ, DE, DK, DM, DZ, EC, EE, ES, FI, GB, GD, GE, GH,
GM, HR, HU, ID, IL, IN, IS, JP, KE, KG, KP, KR, KZ, LC,
LK, LR, LS, LT, LU, LV, MA, MD, MG, MK, MN, MW,
MX, MZ, NO, NZ, PH, PL, PT, RO, RU, SD, SE, SG, SI,
SK, SL, TJ, TM, TR, TT, TZ, UA, UG, US, UZ, VN, YU,
ZA, ZW.(84) Designated States (regional): ARIPO patent (GH, GM,
KE, LS, MW, MZ, SD, SL, SZ, TZ, UG, ZW), Eurasian
patent (AM, AZ, BY, KG, KZ, MD, RU, TJ, TM), European
patent (AT, BE, CH, CY, DE, DK, ES, FI, FR, GB, GR, IE,
IT, LU, MC, NL, PT, SE, TR), OAPI patent (BF, BJ, CF,
CG, CI, CM, GA, GN, GQ, GW, ML, MR, NE, SN, TD,
TG).

Published:

il search report and to be republished
report

Quality Control Patent

- This application was filed in the US on 13 October 2000
- We abandoned the application when the company run out of money
- In hindsight this would have been the most valuable asset of the company



Costs of Acquiring and Maintaining a Patent

- Like any asset, acquiring patents costs money
 - Patent attorney for writing the patent specs and claims
 - Filing costs: filing must be done individually in any country you would like to enforce the patent
 - Translation and attorney fees as well
 - Maintenance fees: every year or every few years one needs to pay a fee to keep the patent current
- These add up quickly: we estimated that each patent over its lifetime of 20 years costs \$300 - \$500,000
- Ericsson's patent portfolio of 20,000+ is a major operational cost

How Much Is a Patent Worth

- Well it depends
 - On the context and the market size
 - How well it is written
- And: “There is no such thing as a perfect patent”
- Given enough time and money any patent can be defeated in courts
 - Is it applicable to the product?
 - Is it novel?
 - This is what happened in the Apple vs Samsung patent litigation



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Apple v Samsung

- When the iPhone came out, Samsung was one of the major mobile phone manufacturers
- They soon started developing something similar to iPhone
 - In view of Apple too similar, and claimed their design patents had been infringed
 - Samsung counter-sued, claiming their 3G technology patents had been infringed
 - The litigation between the two companies in eight jurisdictions (including Australia) went on for some five years (and cost several billion dollars?)
 - I was an expert witness for Apple over 2011-13, examining Samsung's case.



Patent Imperfection

- Samsung had been granted a patent for power control in Australia
- They claimed Apple had not licensed this patent, and that all 3G phone needed to have it.
- In my opinion the patent as written wasn't covering the 3G standards
 - It intended to, but didn't
 - Including Claim 4...

THE CLAIMS DEFINING THE INVENTION ARE AS FOLLOWS:

1. A method of transmitting data of a first channel not supporting Hybrid Automatic Retransmission Request (HARQ) and a second channel supporting the HARQ in a mobile telecommunication system which supports an enhanced uplink service, the method comprising the steps of:
 - determining transmit power factors for the channels and determining if total transmit power required for transmission of the channels exceeds the maximum allowed power;
 - scaling-down the transmit power factor for the second channel if the total transmit power exceeds the maximum allowed power; and
 - transmitting data through the first and second channels using the scaled-down transmit power factor for the second channel and the transmit power factor for the first channel.
2. The method as claimed in claim 1, wherein the scaling step is performed on a slot-by-slot basis.
3. The method as claimed in claim 1, wherein the total transmit power is determined based on the transmit power factors for the first and second channels and a Transmit Power Control (TPC) command issued by the system.
4. The method as claimed in claim 1, further comprising the step of equally reducing transmit power factors corresponding to the other channels comprising the first channel when the transmit power factor for the second channel is scaled down below a predetermined minimum value.

A Career in IP Law

- Most of the lawyers I worked with at Freehills had engineering background
 - They knew the underlying technologies well
 - I *trust* the patent applications they composed wouldn't have the mistakes Samsung patents had
- A patent attorney not familiar with the subject matter makes these mistakes
 - It can't be helped – translating a technology into written word is not straightforward. Mistakes are made, and inventors cannot concentrate to find them
- Having the technical background helps you do a lot better.
 - It can be fun, it is very rewarding, and a great intellectual exercise

Conclusion

- Patents are an important part of product development strategy
 - For defensive and offensive purposes
 - But you need to have the money – which basically rules out small companies
 - However, you still should file, so you are not sued by a competitor...
- Most companies won't bother with going to courts to sue
 - It costs too much
 - The outcome is uncertain
 - The company with deeper pockets can drag the court case and bankrupt you
 - Still, they may settle out of court and pay you a decent royalty...

Summary

- Intellectual property is an important intangible asset
- Most companies need to carefully develop an IP strategy
- Technology companies who develop new products are in a strong place to generate patentable ideas
 - Whether these are applied for depends on
 - Do we have money to initiate the process
 - If patents are granted, how much revenue they could produce
 - Do benefits outweigh costs significantly?